

CFD simulation of flow fields and pollutant dispersion around a cubic building considering the effect of plume buoyancies

Huixin Ma^a, Xuanyi Zhou^{a,*}, Yoshihide Tominaga^b, Ming Gu^a

^a State Key Laboratory of Disaster Reduction in Civil Engineering, Tongji University, 200092, Shanghai, China

^b Department of Architecture and Building Engineering, Niigata Institute of Technology, 1719 Fujiihashi, Kashiwazaki, Niigata, 945-1195, Japan

ARTICLE INFO

Keywords:

Pollutant dispersion
Large-eddy simulation (LES)
Plume buoyancy
Cubic building
Concentration fluctuation

ABSTRACT

Plume buoyancy is one of the most crucial driving forces of gas pollutant dispersion in weak wind regions such as the recirculation region. In this study, steady Reynolds-averaged Navier-Stokes (SRANS) simulation and large eddy simulation (LES) are conducted simultaneously to investigate the flow and concentration fields around a cubic building, in which the positive and negative buoyancies produced by the density difference between pollutants and the ambient air are considered at the same time. The performances of SRANS and LES are quantitatively evaluated by comparing the simulated results with the data from wind tunnel tests. In general, LES provides more accurate predictions for the time-averaged concentration fields, especially for the cases of neutral gas and heavy gas. Also, by studying the effect of different plume buoyant forces, the research finds that the increased vertical turbulent flux of light gas in the recirculation region behind the building enhances the dilution of pollutant. However, the heavy gas has an adverse effect. It is also found that the positive plume buoyancy raises the concentration fluctuation above the pollutant source considerably, while the negative plume buoyancy has a certain inhibitory effect on it.

1. Introduction

In recent years, with the continuous advances in urbanization and industrialization, environmental pollution has raised wide concerns among the public. The release and dispersion of gas pollutants around buildings, which may threaten the health of urban population and do harms to the environment, have aroused huge interests of scientific researchers. Reliable prediction of pollutant dispersion, including the prediction of averaged and instantaneous concentration levels, would facilitate the making of effective counter-measures against the dispersion of hazardous gas. The diffusion of pollutants around buildings is subject to many factors including physical properties of diffused gases, atmospheric stability, the morphology of surrounding buildings, etc. [1–8]. In weak wind regions such as recirculation regions behind the building, the plume buoyancy produced by the density difference between the surrounding air and pollutants is an important factor driving the diffusion of contaminants.

A released fluid containing both momentum and buoyancy is usually called “buoyant jet”. The flow development of buoyant jet is influenced by the combination of jet exit and ambient flow [9]. The buoyant jet in a cross flow is one of the most common and practical scenarios for the

release of gas phase and liquid phase contaminants, and great efforts have been made to it by previous researchers [10–13]. Since buoyant gas diffusion in the urban environment is affected by the complex flow structure around the building, it has certain particularity as a buoyant jet in cross flow. Positive buoyancy is the upward force of lighter-than-air gas, while negative buoyancy is the downward force of heavier-than-air gas. Both are instrumental to the flow fields and the concentration distribution of pollutants [14,15]. The pollutant affected by positive buoyancy is usually associated with high exit temperature, a typical example is the heated plume emitted from the industrial chimney. Some harmful gases have a density difference with air even at room temperature, such as sulfur dioxide (SO₂), which is heavier than air. Once leakage occurs in an emergency, a great threat will be posed to the health of people.

Wind tunnel experiments have been widely used in the studies of near-field pollutant dispersion. However, restricted by similarity criteria, they have encountered problems when being applied to the study of turbulent diffusion processes considering buoyancy effects [16, 17]. For instance, achieving the similarity of Froude number (Fr) normally demands low wind velocity, which could lead to a small Reynolds number, and in this case, it may be difficult to satisfy Reynolds independence [18]. Comparatively speaking, there are fewer wind tunnel

* Corresponding author.

E-mail address: zhouxytj@tongji.edu.cn (X. Zhou).

<https://doi.org/10.1016/j.buildenv.2021.108640>

Received 22 August 2021; Received in revised form 25 November 2021; Accepted 28 November 2021

Available online 2 December 2021

0360-1323/© 2021 Elsevier Ltd. All rights reserved.

Nomenclature			
C	the volume concentration of the tracer gas	V_m	velocity magnitude
c_s	roughness constant	w	vertical velocity
C_s	the Smagorinsky constant	W_s	release velocity of tracer gas
C_0	reference concentration	X	streamwise coordinate
D	molecular diffusion coefficient	Y	spanwise coordinate
D_t	turbulent diffusion coefficient	Z	vertical coordinate
Fr_d	the densimetric Froude number	z_0	aerodynamic roughness length
g	gravitational acceleration	<i>Greek symbols</i>	
H	height of the model building	δ_{ij}	Kronecker delta function
$I_{C,Loc}$	local concentration fluctuation	ε	turbulent dissipation rate
I_u	streamwise turbulence intensity at the inlet boundary	μ	dynamic molecular viscosity
J_i^{SGS}	sub-grid scale mass flux	ν	kinematic molecular viscosity
k	turbulence kinetic energy	ν_{SGS}	sub-grid scale turbulent viscosity
k_s	equivalent sand-grain roughness height	ν_t	kinematic turbulent viscosity
P	pressure	ρ	the density of the mixture
Q_e	the volume flow rate of tracer gas at the source	ρ_a	the density of air
Re	Reynolds number	ρ_s	the density of pollutant
$\bar{s}_{ij}$	mean strain rate tensor	$\Delta\rho$	the density difference between ambient air and pollutant gas
Sc_t	turbulent Schmidt number	τ_{ij}^{SGS}	sub-grid scale Reynolds stress
Sc_{SGS}	sub-grid scale turbulent Schmidt number	τ_w	wall shear stress
t	physical time	φ	the local mass fraction of tracer gas
t^*	non-dimensional sampled time	<i>Variable characteristics</i>	
u_i	velocity component in the direction x_i	$\langle \cdot \rangle$	time-averaged variable
U_H	mean streamwise velocity at the height of building	$\cdot'$	fluctuating variable
u	streamwise velocity	$\bar{\cdot}$	Reynolds averaged variable
u_τ	friction velocity	$\tilde{\cdot}$	resolved-scale variable
v	spanwise velocity		

test studies focusing on the dispersion and transport of non-neutral gases, i.e. light or heavy gases. Lin Chao et al. [19] conducted wind tunnel tests on the dispersion of neutral and light gas around an isolated cubic building. They found that positive buoyancy of light gas can prevent pollutants from getting trapped within the wake region behind the building. A cooperative program, known as Petroleum Environmental Research Forum (PERF) Dispersion Modelling Project 93-16, was carried out in three wind tunnels. This program investigated the transport and dispersion of dense gas plumes over rough surfaces typical of industrial sites. As part of the program, Havens et al. [20] described the downwind and vertical concentration profiles of the gas plume for different surface roughness; Robin et al. [21,22] measured the vertical entrainment velocity of two-dimensional dense gas plumes over fully rough surfaces under neutral and stable boundary conditions, respectively; Briggs et al. [23] summarized the relationship between Richardson number and the ratio of vertical entrainment velocity and buoyancy-induced flow velocity.

Numerical simulation based on computational fluid dynamics (CFD), which is another important tool to investigate pollutant dispersion, could overcome the limitation of similarity criteria on wind tunnel experiments. The most commonly used CFD approaches include two types: Reynolds-averaged Navier-Stokes (RANS) models and Large eddy simulation (LES). RANS solves the mean flow directly and models all scales of turbulence [24], which is known for its economy and reasonable accuracy of different turbulent flows. Olvera et al. [25] performed CFD simulations of neutral and buoyant plumes released within the recirculation cavity behind a cubic building using the renormalization group (RNG) $k-\varepsilon$ model. They analyzed the effects of positive buoyancy and concluded that it could cause considerable flow disturbances inside the wake region, consequently resulting in higher concentrations at the leeward wall of the building and lower concentrations on the ground downstream of the building. Mokhtarzadeh-Dehghan et al. [26] conducted a RANS simulation to investigate the dispersion of heavy gas

released from a ground-level line source. They revealed that with the increase of Richardson number, which means the enhancement of stabilizing effect of negative buoyancy, plume spread further downwind with the depth decreasing. Han et al. [27] studied the dispersion characteristics of heavy gas (SF₆) in an industrial building considering the impact of leakage angle by a chamber experiment and CFD simulation with SST $k-\omega$ model. They concluded that the leakage angle had a greater influence in the lower part of the space than in the upper and middle parts on the concentration distribution of SF₆. Instead of considering the effect of positive or negative buoyancy separately as in the above articles, Tominaga and Stathopoulos [28] took them into account simultaneously. They conducted steady RANS simulations for flow and dispersion fields around an isolated cubic building with light, neutral and heavy gas exhausted behind it. They revealed that the prediction performance of steady RANS simulations with the Boussinesq approximation for the heavy gas was poorer than that for the neutral gas but better than that for the light gas.

Thanks to the growing computing resources, Large-eddy simulation (LES) has attracted increasing attention in recent years [29,30]. LES resolves the large-scale eddies directly thus it only needs to model small-scale eddies which tend to be more universal. Several studies have compared the performances of RANS and LES for pollutants dispersion when the plume buoyancies could be neglected [31–35]. Overall, LES provides a more accurate prediction for flow and concentration fields. However, the literature using LES methods to discuss the effect of plume buoyancies is relatively rare. Hu et al. [36] investigated the fire-induced strong buoyancy effect on the dispersion characteristics of smoke soot and CO gas ejected from side building into urban street canyon under perpendicular wind flow by large eddy simulation (LES). But their study made no quantitative evaluation of the model performance for the predicted concentration and velocity fields. Wingstedt et al. [37] performed large-eddy simulations for the dispersion of neutral and heavy gas released upstream of a simplified urban area. They found that the

plume of heavy gas, compared with that of neutral gas, was wider and shallower and highly affected by the downstream cube array. Lin et al. [38] evaluated the performance of RANS and LES methods in predicting the concentration distribution of high-temperature gas with positive buoyant force but did not fully consider the effect of buoyancy on the flow and concentration fields.

As mentioned earlier, the large-eddy simulations regarding buoyancy gas dispersion are rare, and the discussion about the effect of the buoyant force on concentration field and scalar transport of pollutants is insufficient. In addition, no previous studies compared the performance of RANS and LES in terms of the dispersion of gases with different densities. Therefore, in this study, LES and RANS simulations are performed simultaneously for flow and dispersion fields around an isolated model building, with positive and negative buoyancies being considered. The prediction performances of LES and RANS in terms of mean flow and concentration fields are evaluated by comparing the simulated results with experimental data of Tominaga and Stathopoulos [28]. Then the effect of plume buoyancies is further illustrated using the predicted results of LES, and the characteristics of pollutant dilution, pollutant transport, and concentration fluctuation under the effect of different buoyancies are subsequently analyzed.

2. Outlines of wind-tunnel experiments

The research objective in this paper is the same as that of the wind tunnel experiments carried out by Tominaga et al. [39]. As shown in Fig. 1, a cubic building with the sides $H = 0.2$ m is located in the atmospheric boundary layer. The square pollutant source, which has a side of $0.025H$, is $0.5H$ away from the leeward side of the building. This experiment was conducted in a wind tunnel at the Institute of Industrial Science at the University of Tokyo. Fig. 2 shows the inflow profiles of dimensionless mean streamwise velocity $\langle u \rangle / U_H$ and turbulence intensity $I_u = \sqrt{\langle u'^2 \rangle} / \langle u \rangle$, both of which were measured in the center of the turntable (i.e., the center of building location during the test) in an empty wind tunnel. The inflow wind is perpendicular to the windward side of the model building. The mean streamwise velocity at building height U_H is 0.4 m/s, while the turbulence intensity at the same height is approximately 20%. The Reynolds number $Re = U_H H / \nu$ is 5.7×10^3 . According to the experimental results of Tominaga and Stathopoulos [28], the measured velocity of $Re = 5.7 \times 10^3$ differs very little from that of $Re = 5.7 \times 10^4$.

To investigate the effect of positive and negative plume buoyancy,

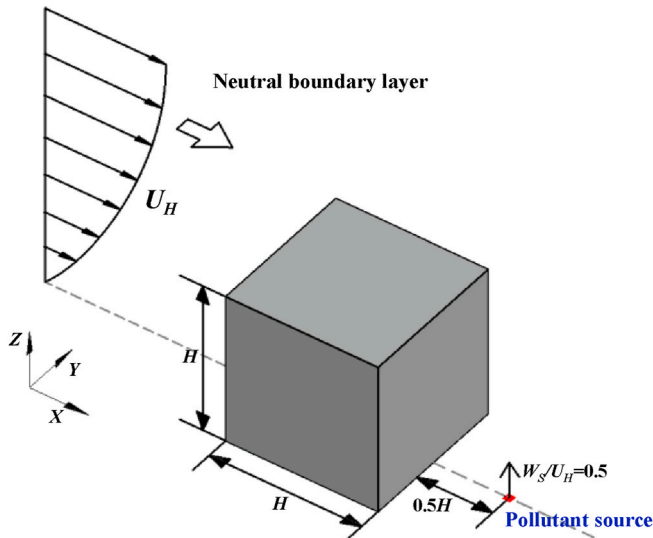


Fig. 1. Schematic diagram of the model building.

three cases are set according to the density difference between the ambient air and tracer gases, as shown in Table 1. The densimetric Froude number Fr_d here is defined as follows

$$Fr_d = \frac{Q_e \Delta \rho}{\rho_a U_H H^2} \quad (1)$$

where ρ_a is the density of air; $\Delta \rho = \rho_a - \rho_s$ is the density difference between air and pollutants.

In this experiment, ethylene (C_2H_4), which has a density similar to that of air, is used as neutral gas. The mixture of ethylene and helium (He), ethylene and sulfur hexafluoride (SF_6) are used as the light and heavy gas respectively. The release velocity of pollutants W_s is set to $0.5 U_H$.

It should be noted that all the wind velocity data in this study, including those from experiments and simulations, are non-dimensionalized using the reference velocity U_H , while the reference concentration C_0 is used for dimensionless processing of volume concentration, defined as follows:

$$C_0 = \frac{Q_e}{H^2 U_H} \quad (2)$$

where $Q_e = 5 \times 10^{-6} m^3/s$ is the volume flow rate of tracer gas at the source, which is defined as the product of the gas source area and the release velocity. The value of C_0 is 312.5 ppm_v in the study. The relationship between volume concentration C and mass fraction φ is expressed as:

$$C = \frac{\varphi \rho_a}{(1 - \varphi) \rho_s + \varphi \rho_a} \quad (3)$$

3. Computational setup

3.1. Numerical methods and solver settings

In this paper, LES and SRANS are used to investigate pollutant diffusion under the effect of different plume buoyancies. The governing equations are solved based on the finite volume method, using the commercial software ANSYS Fluent [40].

3.1.1. SRANS

For the steady RANS methods, the Reynolds-averaged governing equations are shown in Eqs. (4)–(6).

$$\frac{\partial \rho \bar{u}_i}{\partial x_i} = 0 \quad (4)$$

$$\frac{\partial}{\partial x_j} (\rho \bar{u}_i \bar{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} (2\mu \bar{s}_{ij}) - \frac{\partial}{\partial x_j} (\rho \overline{u'_i u'_j}) + \rho g \delta_{i3} \quad (5)$$

$$\frac{\partial}{\partial x_i} (\rho \bar{u}_i \bar{\varphi}) = \frac{\partial}{\partial x_i} \left(\rho D \frac{\partial \bar{\varphi}}{\partial x_i} \right) - \frac{\partial}{\partial x_i} (\rho \overline{u'_i \varphi'}) \quad (6)$$

where $\bar{s}_{ij} = \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right)$ is the mean strain rate tensor.

The Reynolds stress $-\overline{u'_i u'_j}$ and turbulent mass flux $-\overline{u'_i \varphi'}$ which represents the impact of turbulence on mean values are modeled by the Boussinesq hypothesis and standard gradient-diffusion hypothesis respectively, as shown in Eqs. (7)–(8).

$$-\overline{u'_i u'_j} = 2\nu_t \bar{s}_{ij} - \frac{2}{3} k \delta_{ij} \quad (7)$$

$$-\overline{u'_i \varphi'} = \frac{\partial \bar{\varphi}}{\partial x_i} D_t \quad (8)$$

where $D_t = \frac{\nu_t}{Sc_t}$ is the turbulent diffusion coefficient.

Based on the previous research of Tominaga and Stathopoulos [28],

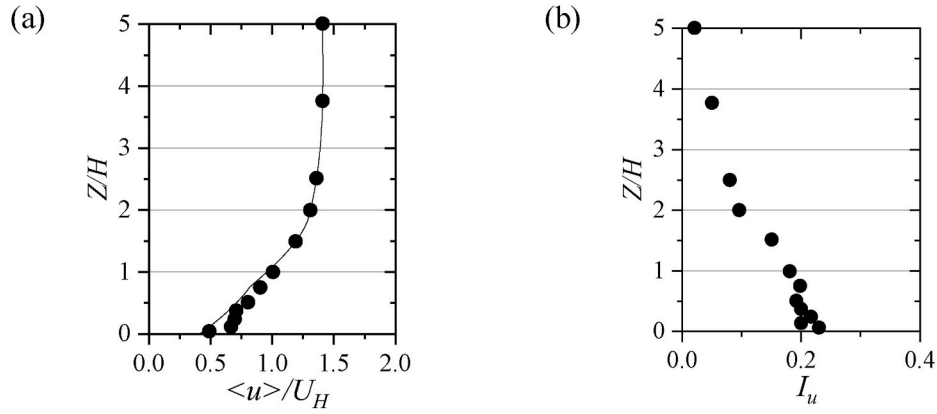


Fig. 2. Inflow profiles of (a) dimensionless mean streamwise velocity and (b) turbulence intensity.

Table 1

Test cases (The same case settings are used as Tominaga et al. [28]).

Case	ρ_s/ρ_a	$\Delta\rho/\rho_a$	Fr_d
Light gas	0.3	0.7	2.2×10^{-4}
Neutral gas	1.0	0	0.0
Heavy gas	1.7	-0.7	-2.2×10^{-4}

the Realizable k- ϵ model, which provides the best agreement with experimental data among the four two-equation turbulence models, is used in our study. Since the standard gradient-diffusion hypothesis (SGDH) is used in RANS to consider the turbulent diffusion of scalar, the turbulent Schmidt number Sc_t plays an important role in the simulation of scalar dispersion. Previous studies [28,38] have examined the sensitivity of Sc_t to the prediction performance of different buoyant gases. As the major aim of this paper is to compare the simulation accuracy between SRANS and LES, the effect of Sc_t is ignored. The same Sc_t value is chosen for all cases, which is 0.7. Calculation convergence is assumed to be obtained when the residual for the mass, momentum, and species transport equations is lower than 10^{-6} .

3.1.2. LES

The filtered governing equations of LES for mass, momentum, and pollutant transport are shown in Eqs. (9)–(11).

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} (\rho \tilde{u}_i) = 0 \quad (9)$$

$$\frac{\partial}{\partial t} (\rho \tilde{u}_i) + \frac{\partial}{\partial x_j} (\rho \tilde{u}_i \tilde{u}_j) = \mu \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} - \frac{\partial \tau_{ij}^{SGS}}{\partial x_j} + \rho g \delta_{i3} \quad (10)$$

$$\frac{\partial}{\partial t} (\rho \tilde{\varphi}) + \frac{\partial}{\partial x_i} (\rho \tilde{u}_i \tilde{\varphi}) = \rho D \frac{\partial^2 \tilde{\varphi}}{\partial x_i \partial x_i} - \frac{\partial \rho J_i^{SGS}}{\partial x_i} \quad (11)$$

The sub-grid scale Reynolds stress $\tau_{ij}^{SGS} = \tilde{u}_i \tilde{u}_j - \tilde{u}_i \tilde{u}_j$ and the sub-grid scale mass flux $J_i^{SGS} = \tilde{u}_i \tilde{c} - \tilde{u}_i \tilde{c}$ are modeled as follows:

$$\tau_{ij}^{SGS} = -\nu_{SGS} \frac{\partial \tilde{u}_i}{\partial x_j} \quad (12)$$

$$J_i^{SGS} = -\frac{\nu_{SGS}}{Sc_{SGS}} \frac{\partial \tilde{\varphi}}{\partial x_i} \quad (13)$$

where ν_{SGS} is the sub-grid scale turbulent viscosity, Sc_{SGS} is the sub-grid scale turbulent Schmidt number.

The Standard Smagorinsky-Lilly model [41] with Smagorinsky constant $C_s = 0.12$ is applied to model ν_{SGS} , which has been proved to be an applicable choice for the simulation of flow and dispersion in built environments [3,32,42]. The value of Sc_{SGS} only has an impact on sub-grid

scale turbulent flux, which is generally one or two orders of magnitude lower than the resolved turbulent flux [33]. Therefore, compared with Sc_t in RANS, Sc_{SGS} in LES has much less influence on the concentration field. Here, Sc_{SGS} is set to 0.7 for all cases, which is a commonly used value for the dispersion around an isolated building [7,8,35,38]. The time step is set to $\Delta t = 0.0015s$ ($0.003 H/U_H$) to make sure that the Courant-Friedrichs-Lewy (CFL) numbers are less than 1.0. The resulting maximum CFL number is 0.93, which occurs in the upper part of the computational domain above the gas source. The calculation before statistical sampling lasts for 30s ($60 H/U_H$), while the sampling process continues for 180s ($360 H/U_H$), which is considered long enough for the statistical convergence of time-averaged values according to a validation shown in Section 3.5.

The discretization schemes for both SRANS and LES are summarized in Table 2. The LES simulations are performed on the supercomputer with a 64-core node, while the RANS simulations are conducted by a high-performance computer with 6 cores. The computational time of LES and RANS for each case is 136h and 25h, respectively.

3.1.3. Buoyancy model

In this study, the density of mixture in governing equations is treated as a variable value, which relates to the density of pollutant and air by the volume weight mixing law

$$\rho = \frac{1}{\varphi \rho_s^{-1} + (1 - \varphi) \rho_a^{-1}} \quad (14)$$

This method is also successfully used for the simulation of dense gas dispersion over a simplified urban area by Wingstedt et al. [37]. Incompressible flow with the Boussinesq approximation is another method commonly used for flow with density changes. It treats density as a constant value in all solved equations, except for the buoyancy term in the momentum equation. Tominaga and Stathopoulos [28] verified the effectiveness of SRANS with Boussinesq approximation for the current release cases of buoyant gas. However, the Boussinesq approximation is accurate only when the changes of density are small ($|\Delta\rho/\rho_a| \ll 1$) [45], and since this is not satisfied in the current simulation, the more general formulation detailed above is used in this study.

3.2. Computational domain

SRANS and LES have an equal computational domain of $21.0H(X) \times 9.7H(Y) \times 5.4H(Z)$, as shown in Fig. 3(a). To limit the streamwise gradients of inflow profiles, the distance between the inlet boundary and the windward wall of the building is set to $5H$ [43]. The downstream length of the computational domain is $15H$. The blockage ratio is less than 3%, satisfying the requirements of the Architectural Institute of Japan (AIJ) guidelines [46]. In this study, the origin of coordinates is always located in the building center as shown in Fig. 3(a).

Table 2
Numerical analysis and boundary conditions for all simulation cases.

Simulation model	RANS	LES
Simulation domain	21H (X) × 9.7H (Y) × 5.4H (Z)	
Simulation mesh	Hexahedral, 1.45×10^6	
Turbulence model	Realizable k-ε model	Standard Smagorinsky-Lilly model
Velocity pressure coupling/time marching	SIMPLE	SIMPLEC, bounded second order implicit scheme
Discretization scheme for momentum and concentration equation	Second-order upwind scheme	Bounded central differencing scheme
Discretization scheme for pressure	Second order scheme	Second order scheme
Inlet boundary condition	Time-averaged streamwise velocity from the experiment, turbulence kinetic energy $k(Z) = a(I_u(Z)u(Z))^2$, turbulent dissipation rate $P_k = \varepsilon$	Vortex method with the number of vortices selected as 190, time-averaged streamwise velocity from the experiment, turbulence kinetic energy $k(Z) = a(I_u(Z)u(Z))^2$, turbulent dissipation rate $P_k = \varepsilon$
Gas source boundary condition	Velocity inlet with exhaust velocity $W_e = 0.5U_H = 0.2\text{m/s}$, mass fraction of tracer gas $\varphi = 1$, and no turbulence.	
Outlet boundary condition	Zero gradient	
Upper and side boundary	Symmetry	
Ground and building surfaces	Non-slip wall and standard wall functions with $k_s = 0.973z_0/c_s$ [43] ($z_0 = 0.0004\text{m}$) and $c_s = 0.5$.	Non-slip wall with the near-wall treatment in ANSYS Fluent using the blending function [44]
Time step size		0.0015s
Flow time		210s
Time-averaged flow time		180s
Non-dimensional time units		360
$t^* = t \times U_H / H$		
Courant-Friedrichs-Lewy (CFL) number		≤0.93

3.3. Boundary conditions

The boundary conditions of SRANS and LES simulations are summarized in Table 2. The fitting curve of the mean inflow velocity from wind tunnel experiments is used as the inlet boundary condition for both

SRANS and LES simulations, shown as the solid line in Fig. 2(a). The turbulence kinetic energy k is calculated by Eq. (15) according to the recommendation of AIJ guidelines [46], where the constant a is set to 1. The values of turbulent dissipation rate ε are obtained based on the local equilibrium $P_k = \varepsilon$ in the k equation, where P_k is the production term. For LES, the velocity fluctuation at the inlet boundary is generated by the vortex method [47,48] with a vortex number of 190 [33]. According to previous researchers [4,8,49–51], vortex method presented a great performance for flow and dispersion around buildings.

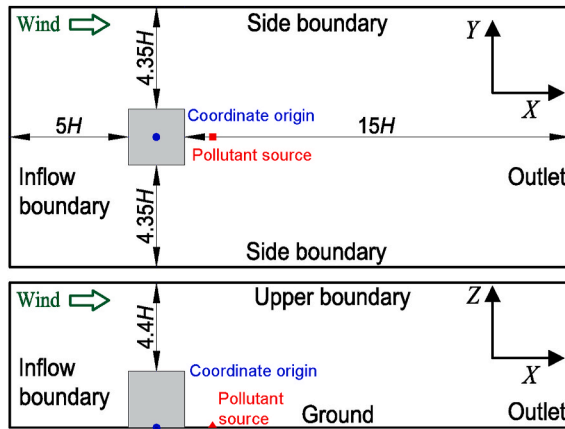
$$k(Z) = a(I_u(Z)\langle u(Z) \rangle)^2 \quad (15)$$

3.4. Grid sensitivity analysis

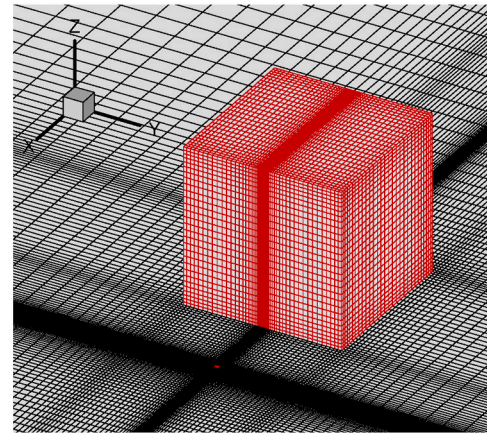
Grid sensitivity analysis is conducted based on LES with three different grids. The details of them are summarized in Table 3. The height of the first layer grid is $H/25$, $H/40$ and $H/50$ for coarse, medium and fine grid, making the Y^+ ($Y^+ = \rho u_\tau y / \mu$, where $u_\tau = \sqrt{\tau_w / \rho}$) have an average value of 7, 5 and 4 for these three grids, respectively. The grid distribution around the building and pollutant source for the medium scheme is shown in Fig. 3(b). Fig. 4 compares the time-averaged dimensionless streamwise velocity for neutral gas case obtained from the three different grids along three vertical profiles at plane $Y/H = 0$. No significant discrepancies were observed under the building height ($Z/H < 1.0$) at each location. However, the results of coarse grid exhibited a small deviation from that of the other grids in the higher part ($Z/H > 1.0$), especially along line $X/H = 1.0$, while the results of medium and fine grids are similar. Therefore, the medium grid system is retained for the following simulation. It should be noted that this grid sensitivity study is carried out based on LES method, and the same grid is

Table 3
Characteristics of the computational grids.

	Coarse grid	Medium grid	Fine grid
Grid discretization	136(X) × 87(Y) × 38(Z)	198(X) × 117(Y) × 65(Z)	274(X) × 158(Y) × 71(Z)
Total number of cells	441693	1449931	2991812
Stretching ratio	Less than 1.2	Less than 1.1	Less than 1.1
Number of grids along the building	19(X) × 47(Y) × 19(Z)	26(X) × 53(Y) × 26(Z)	32(X) × 80(Y) × 32(Z)
Number of grids along the pollutant source	3(X) × 3(Y)	5(X) × 5(Y)	8(X) × 8(Y)
Height of the first layer of grid near floor	H/25	H/40	H/50
Average Y^+ value	7	5	4



(a) Computational domain



(b) Grid system

Fig. 3. Schematic diagram of (a) computational domain and (b) grid system near the model building (the total number of grids is around 1.45 million).

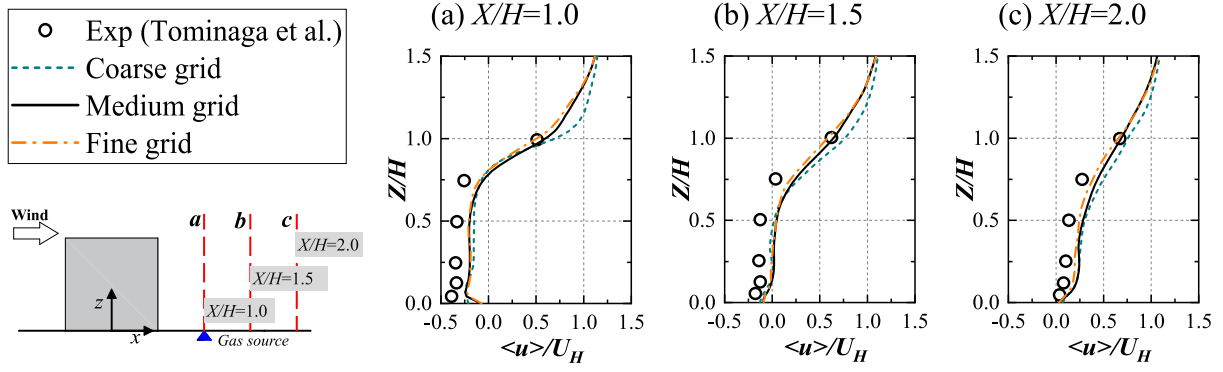


Fig. 4. Grid sensitivity analysis results: vertical profiles of time-averaged dimensionless streamwise velocity $\langle u \rangle / U_H$ for the neutral gas case at plane $Y/H = 0$.

used for SRANS simulation, which is considered absolutely adequate.

To obtain LES simulation results with more credibility, the paper considers the index of quality (LES_{IQ}), which is defined as the ratio of the resolved turbulence kinetic energy (rTKE) to the total turbulence kinetic energy (tTKE) [52]. The LES_{IQ} is estimated according to the following formula

$$LES_{IQ} = \frac{k_{resolved}}{k_{total}} = \frac{1}{1 + 0.05 \left(\frac{\nu + \nu_{SGS}}{\nu} \right)^{0.53}} \quad (16)$$

As recommended by many previous researchers [53–55], in well-resolved regions, the fraction of rTKE should be more than 80%. Table 4 gives the average value of LES_{IQ} obtained by medium grid at six different lines near the building in the plane $Y/H = 0$, with all results of around 95%. Therefore, it could be concluded that the medium grid is fine enough to resolve most of the turbulence kinetic energy.

3.5. Monitoring of statistical convergence

Fig. 5 depicts the changing process of dimensionless time-averaged vertical velocity and concentration at the monitoring point as a function of the sampling time. The monitoring point is located at $X/H = 1.0$, $Y/H = 0$, $Z/H = 0.5$, as shown in the schematic diagram in Fig. 5. What's worth noticing is that to achieve the statistical convergence, the sampling time for each case is 16 times longer than flow-through time ($T = L/U_H$ with L being the streamwise length of the computational domain and U_H being the reference velocity). As shown in Fig. 5 (a)(b), the time-averaged vertical velocity and concentration at the monitoring point generally remain stable after 60 s of sampling, which indicates the sampling time in the study is ample, despite some small variations.

4. Comparison between RANS and LES

4.1. Mean flow field

Fig. 6 compares the predicted results of time-averaged streamwise velocity ($\langle u \rangle / U_H$) behind the building with the experiment data

quantitatively. It should be noted that this comparison is conducted only for the neutral gas case due to the limitation of experimental data. Both RANS and LES successfully reproduce the distribution trend of $\langle u \rangle / U_H$ at the wake region behind the building. At $X/H = 1.0$, the results of these two methods are similar to each other, while a certain underestimation of the reverse flow is observed for both of them. A strong reverse flow is predicted by RANS at $X/H = 1.5$ and $X/H = 2.0$, which means that it overestimates the length of the recirculation region behind the building. Comparatively, LES provides more accurate predictions of the reattachment length than RANS, as is indicated in previous studies [32,56]. In this study, a low inflow wind velocity with $U_H = 0.4 \text{ m/s}$ is set to highlight the effect of plume buoyancy on weak wind regions such as the recirculation region, causing some difficulties to both measurements in wind tunnels and prediction by numerical methods of wind velocity.

Fig. 7 shows the time-averaged streamlines of LES and RANS at the vertical plane ($Y/H = 0$) and horizontal plane ($Z/H = 0.5$). The two planes are contoured by vertical velocity ($\langle w \rangle / U_H$) and velocity magnitude ($\langle V_m \rangle / U_H$) respectively. The streamlines visually demonstrate the reattachment length and the size of the recirculation region. For the neutral gas case, the reattachment length predicted by LES is about $1.1H$, which is very similar to the wind tunnel results for cubic building of $1.05H$ [38] and $1.33H$ [32], and also the full-scale measurement results of $1.4H$ [57]. However, RANS exhibits an overestimated length of the recirculation region of $2H$. At the same time, the separation and reattachment above the roof and along the building side are also successfully reproduced by LES. Fig. 8 provides the distribution of turbulence kinetic energy k at plane $Y/H = 0$. For LES, the resolved k is calculated by $k = 1/2(\sigma_u^2 + \sigma_v^2 + \sigma_w^2)$, where σ_i is the standard deviation of resolved velocity component in the i direction. It can be seen that the positive buoyancy of the light gas increases the value of k near the pollutant source, while the negative buoyancy of the heavy gas suppresses it to some degree. This is also observed by Tominaga and Stathopoulos [28] in wind tunnel tests. Both the results of SRANS and LES reflect this tendency successfully, but the value of k predicted by SRANS behind the building is obviously smaller than that given by LES, which is mainly attributed to the fact that SRANS cannot reproduce the

Table 4
LES quality index for medium grid.

Location	a $X/H = -1.5$	b $X/H = -1.0$	c $X/H = -0.6$
Average LES_{IQ}	95.00%	95.00%	95.06%
Location	d $X/H = 0.6$	e $X/H = 1.0$	f $X/H = 1.5$
Average LES_{IQ}	95.03%	95.16%	95.01%

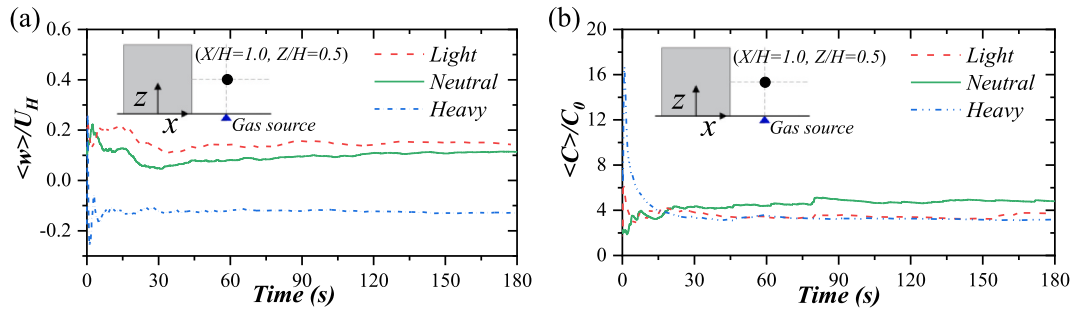


Fig. 5. Convergence monitoring for three cases: (a) dimensionless vertical velocity $\langle w \rangle / U_H$ and (b) dimensionless concentration $\langle C \rangle / C_0$ as a function of sampling time.

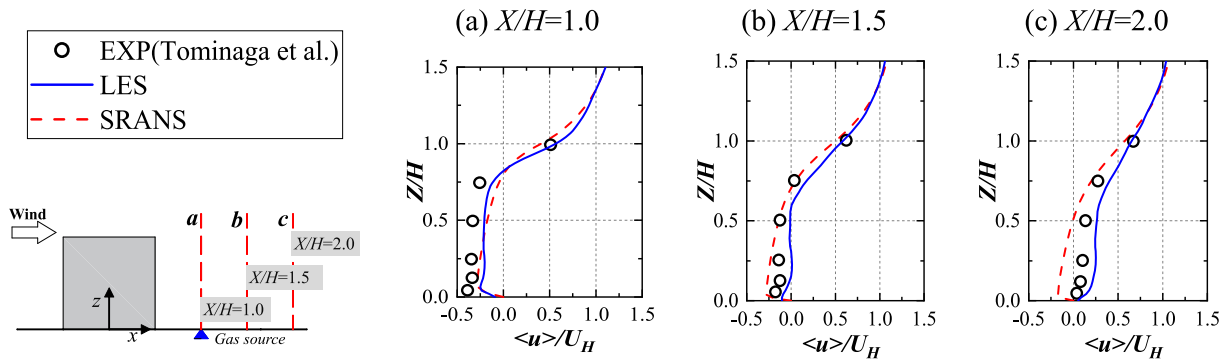


Fig. 6. Profiles of time-averaged dimensionless streamwise velocity $\langle u \rangle / U_H$ for neutral gas case behind the building at vertical plane $Y/H = 0$.

periodic fluctuation caused by vortex shedding behind the building. This phenomenon is closely related to the prediction of reattachment length by these two methods. Compared with SRANS, in LES simulation, a large lateral momentum diffusion is generated behind the building due to the reproduction of vortex shedding, resulting in smaller reverse flows and reattachment lengths that are closer to the experimental results.

In terms of the simulated results of LES (Fig. 7(a)), significant differences are observed in the structure of the wake vortex with the impact of different buoyancies. For the light gas case, under the effect of upward positive buoyant force, a strong updraft appears near the pollutant source and the leeward side of the building. As a result, the streamlines in the recirculation region present an obvious upwash trend and a smaller vortex structure is generated. For the heavy gas case, the raising of the airflow in the recirculation region is suppressed by the downward negative buoyant force, leading to a larger vortex structure with the streamlines under the vortex parallel to the ground. By contrast, there is almost no difference between the vortex structures predicted by RANS under the effect of different buoyancies.

4.2. Mean concentration field

Fig. 9 illustrates the contours of time-averaged concentration predicted by CFD and obtained from wind tunnel experiment at the vertical plane ($Y/H = 0$) and horizontal plane ($Z/H = 0.5$). On the whole, as the pollutant source is located in the recirculation region behind the building, pollutants with different buoyancies are transported towards the leeward wall of the building under the effect of reverse flow produced by the wake vortex after being released from the source. Therefore, a high concentration is distributed between the leeward side of the building and the source of gas. The neutral gas case provides a reference for the other two cases to highlight the effect of positive and negative buoyancies. The concentration distribution behind the building simulated by both LES and RANS are generally consistent with that of the experiments, although overestimation is observed at $Z/H > 0.5$ behind the gas source for RANS. For the light gas case, affecting by the upward

buoyancy force, a clear angle between the plume centerline and the ground is found in the result of LES, and at the same time, a relatively low concentration region is shown under the plume (Fig. 9(b)). These characteristics are in line with the results from the wind tunnel experiments by Lin et al. [19], while RANS fails to reproduce them. Comparing with experimental results, both LES and RANS successfully reproduce the distribution characteristics of light gas, which has a higher concentration and transports further along streamwise at the height of the building roof. However, there is an obvious overestimation in RANS while LES provides a more appropriate prediction. For the heavy case, pollutants accumulate in the lower part ($Z/H < 0.5$) and transport downstream under the effect of downward buoyancy force. The result of LES shows a close agreement with that of the experiment in the plane $Y/H = 0$, while RANS provides a much higher concentration at the height of the building in the wake region, which is quite different from the experimental data. To be consistent with the experimental data shown in the following figures, the height of $Z/H = 0.5$ is selected to show the horizontal plane. No significant discrepancy of the horizontal concentration distribution of pollutants with different buoyancies is observed at this height, although compared with LES, RANS provides a higher concentration in the wake region.

To validate the simulation accuracy of LES and RANS under the effect of different plume buoyancies, a comparison is made between simulated and measured concentration along three vertical lines in the center plane $Y/H = 0$ and three horizontal lines in the cross-plane $Z/H = 0.5$, which are set along the downwind direction: $X/H = 1.0$, $X/H = 1.5$, $X/H = 2.0$. The percentage deviation at each measuring point is defined as the absolute difference between simulated results and experimental data divided by the experimental data. Fig. 10 depicts the concentration distribution of the light gas case. Although LES and RANS can largely reproduce the vertical diffusion trend of it, RANS overpredicts the concentration of contaminants near the ground level at $X/H = 1.5$ (line b) and $X/H = 2.0$ (line c). For line b, the average percentage deviations of LES and RANS are 30.97% and 67.43%, respectively. A relatively great consistency between CFD results and wind tunnel data can be seen

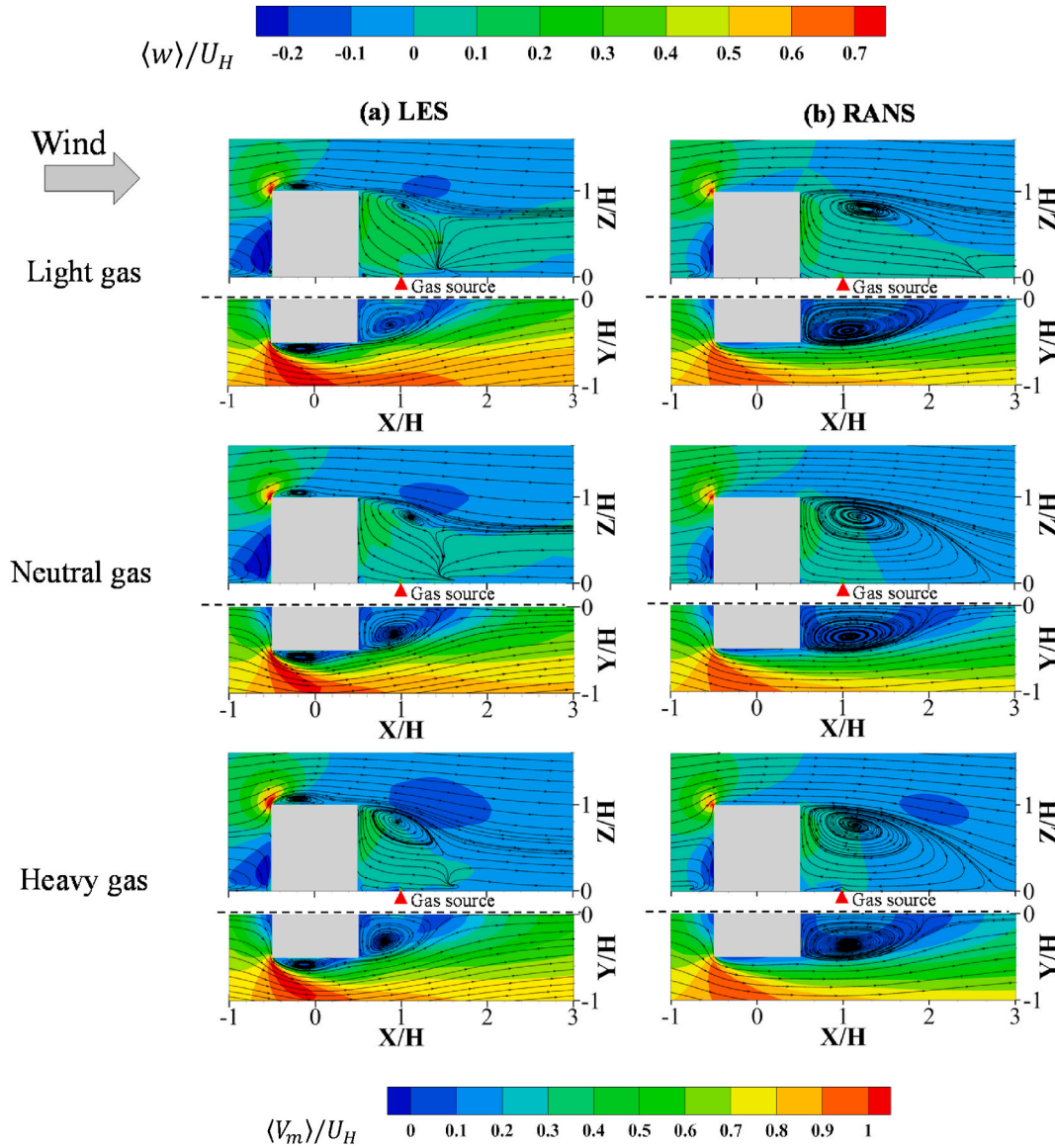


Fig. 7. The distribution of time-averaged streamlines at vertical plane $Y/H = 0$ (contoured by the time-averaged dimensionless vertical velocity $\langle w \rangle / U_H$) and horizontal plane $Z/H = 0.5$ (contoured by the time-averaged dimensionless velocity magnitude $\langle V_m \rangle / U_H$).

for line c, with the average percentage deviation of 10.35% for LES and 36.93% for RANS. In horizontal plane, the total average percentage deviation is 27.33% for LES and 29.60% for RANS. However, underestimations of the horizontal diffusion are found in LES, especially for line f. Since the positive buoyancy of light gas increases the turbulence kinetic energy near the pollutant source, a possible reason for the underestimation of horizontal diffusion is that the current grid resolution near the source is not higher enough to capture the initial dispersion generated by small eddies. That should be investigated in future studies. Fig. 11 and Fig. 12 indicate the vertical and horizontal concentration profiles of neutral and heavy gas cases, respectively. Results from LES exhibit satisfactory agreement with experiment findings, with the total average percentage deviations for neutral and heavy gas cases being 18.03% and 25.91%, respectively. However, for the results of RANS, overestimations of the concentration transport in the horizontal plane are observed, which are particularly noticeable for the heavy gas case. These overestimations account largely for the poor performance of RANS, which has a total average percentage deviation for neutral and heavy gas case of 50.63% and 101.02%, respectively. Discrepancies between the simulated and wind tunnel data are observed for three cases

in the lower part ($Z/H \leq 0.25$) of the vertical line above the pollutant source ($X/H = 1.0$), especially for light and heavy gas cases. A possible reason that accounts for the discrepancies is the uncertainty of wind tunnel tests, as the close distance to the floor presents some difficulties in the measuring of concentration fields. Similar simulated results of the concentration distribution in the lower part were obtained by Tominaga and Stathopoulos [28] for the same wind tunnel tests, which further indicates that some experimental errors might exist.

To describe the prediction accuracy of CFD methods (LES and RANS) with the effect of different plume buoyancies in a more intuitive way, three validation metrics are used here: the fraction of the prediction within a factor of 2 of the observation (FAC2), fractional bias (FB), and normalized mean square error (NMSE) [58]. The metrics can be calculated as follows:

$$FAC2 = \frac{1}{N} \sum_{i=1}^N n_i \text{ with } n_i = \begin{cases} 1 & \text{for } 0.5 \leq \frac{P_i}{O_i} \leq 2 \\ 1 & \text{for } O_i \leq W \text{ and } P_i \leq W \\ 0 & \text{else} \end{cases} \quad (17)$$

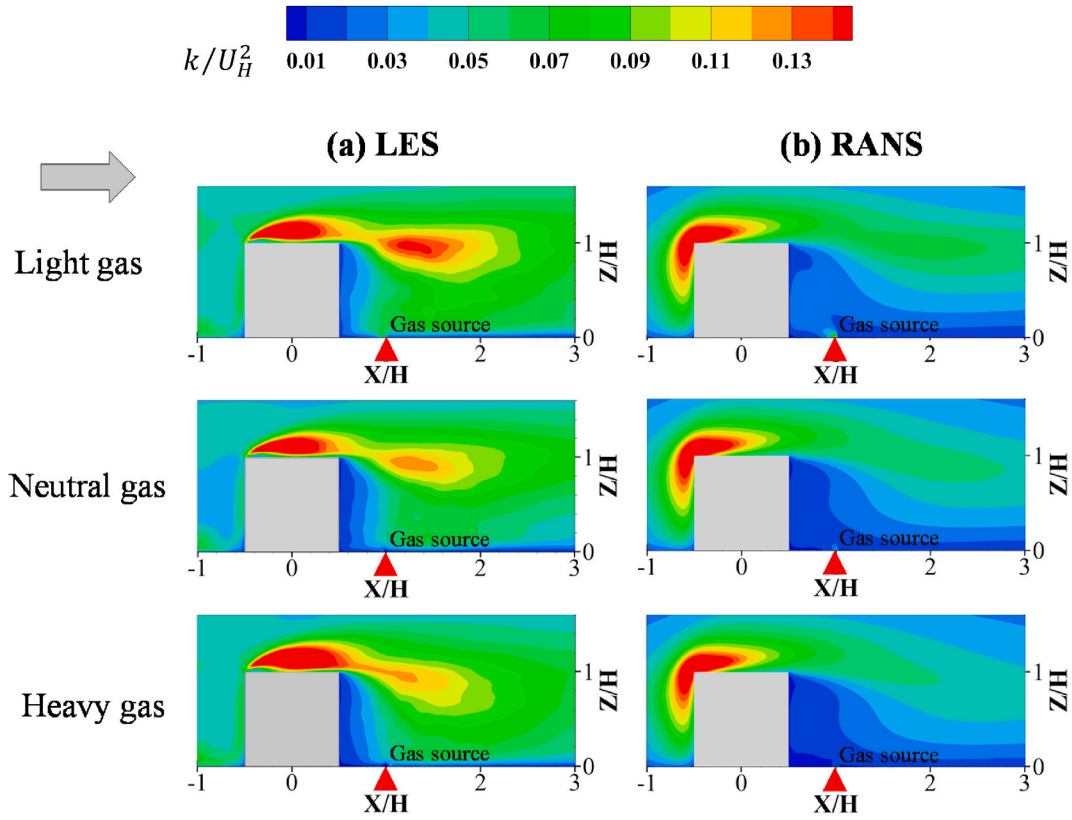


Fig. 8. The distribution of dimensionless turbulence kinetic energy k/U_H^2 at plane $Y/H = 0$.

$$FB = \frac{[O] - [P]}{0.5([O] + [P])} \quad (18)$$

$$NMSE = \frac{[(O_i - P_i)^2]}{[O][P]} \quad (19)$$

where O and P are the observed (measured) and predicted (simulated) values of concentration at i_{th} sample point respectively, and N is the number of data points. The allowed absolute diffidence W is set to 0.05 for FAC2. The square brackets indicate that the variable is averaged over the whole dataset.

Table 5 lists the values of the validation metrics for RANS and LES. All the data points from the three cases shown in the above figures (Figs. 10, Fig.11, and Fig. 12) are used to calculate the validation metrics. The ideal value which means perfect consistency between measured data and simulated results is 1 for FAC2 and 0 for FB and NMSE. According to studies of previous researchers [58,59], the acceptance range of three metrics are as follows: $FAC2 > 0.5$, $|FB| < 0.3$, and $NMSE < 4$. As shown in Table 5, most metric values for light, neutral and heavy gas of both RANS and LES are within the acceptance range. Overall, LES performs better than RANS for most metrics, with some exceptional values which indicate the poor performance of LES being underlined in Table 5. The relatively poorer performance of LES mainly relates to the under-prediction of horizontal distribution of light gas, as shown in Fig. 10. Considering the effect of different plume buoyancies, the performance of LES for neutral gas is better than that for light and heavy gases with the value of FAC2 in both vertical and horizontal planes at 1. In addition, LES performs better for heavy gas than light gas in terms of FAC2 in both the vertical plane (0.95 for heavy gas and 0.92 for light gas) and horizontal plane (0.94 for heavy gas and 0.79 for light gas). As a result, the conclusion can be arrived at that, for the prediction of time-averaged concentration, LES performs the best for the neutral gas case, followed by the heavy gas case and finally the light gas case.

5. Detailed analysis of the effects of plume buoyancy

As discussed above in section 4.2, for the time-averaged concentration under the effect of different plume buoyancies, LES provides more accurate results than RANS in general. In addition, important fluctuation information could be gained from it. Therefore, in this section (from Fig. 13 to Fig. 17), only the results of LES will be used to further discuss the effect of plume buoyancies on the characteristics of pollutant dilution, pollutant transport, and concentration fluctuation.

5.1. Impact of plume buoyancy on pollutant dilution behind the building

Fig. 13 depicts the contaminants distribution near the leeward side of the building. Comparing to the neutral gas case (Fig. 13 (b)), the light gas (Fig. 13 (a)) has a relatively even distribution in the leeward wall, with the high concentration staying around the center of the wall, where plume impinging occurs. However, a large concentration gradient from the bottom to the top of the leeward side is observed in the heavy gas case (Fig. 13 (c)). Meanwhile, the pollutants at the bottom part of the building diffuse across the wind substantially. On the leeward side of the building, the peak values of time-averaged concentration ($\langle C \rangle / C_0$) for the light gas case and the neutral gas case are about 22 and 41 respectively, while that of the heavy case already exceeds 80.

Fig. 14 presents the plane-averaged dimensionless time-averaged concentration in several cross-sections behind the building along the streamwise direction; the width and height of each section are equal to that of the leeward wall of the building. In the recirculation region ($0.5 < X/H < 1.6$), significant differences are observed among the three cases. The plane-averaged concentration of heavy gas is considerably higher than that of neutral gas and light gas, especially at the plane of pollution source ($X/H = 1.0$). In the downstream of the recirculation region ($X/H > 1.6$), the difference between the three cases decreases downstream gradually, with the light gas maintaining the lowest

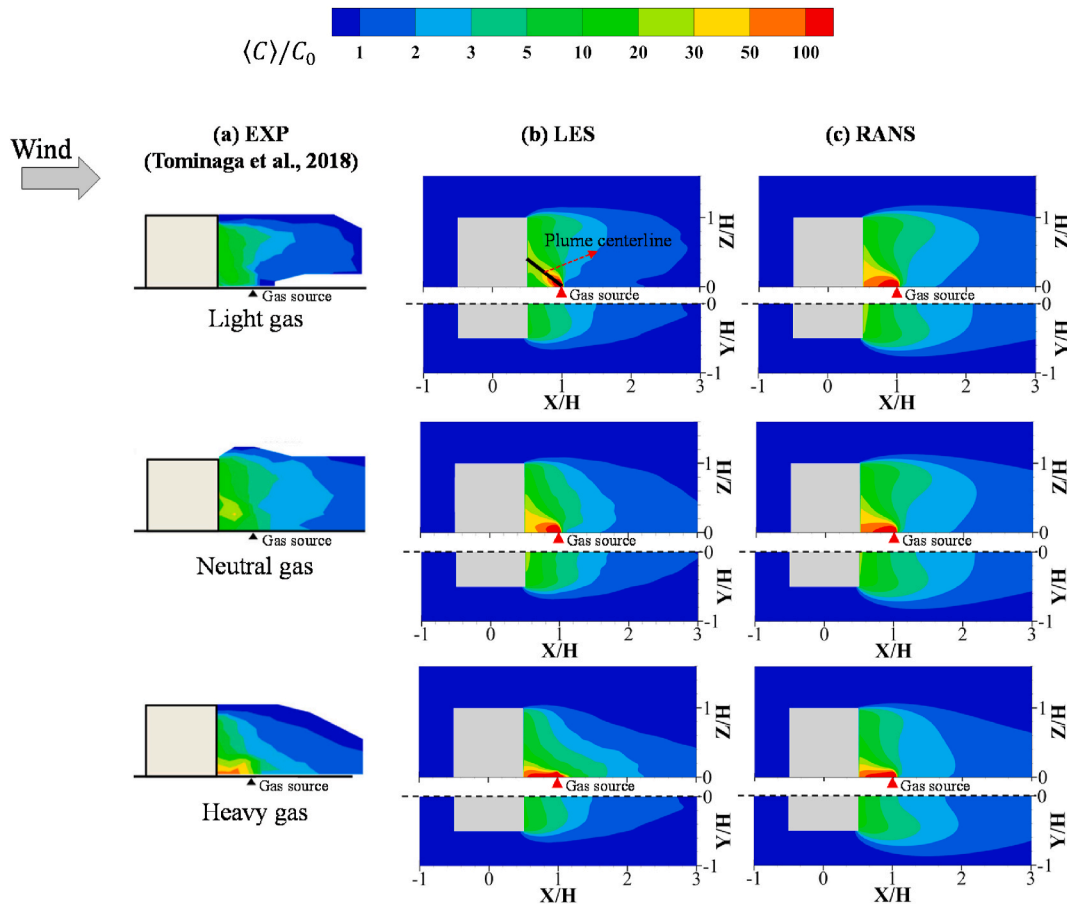


Fig. 9. Contours of time-averaged dimensionless concentration $\langle C \rangle / C_0$ at vertical plane $Y/H = 0$ and horizontal plane $Z/H = 0.5$.

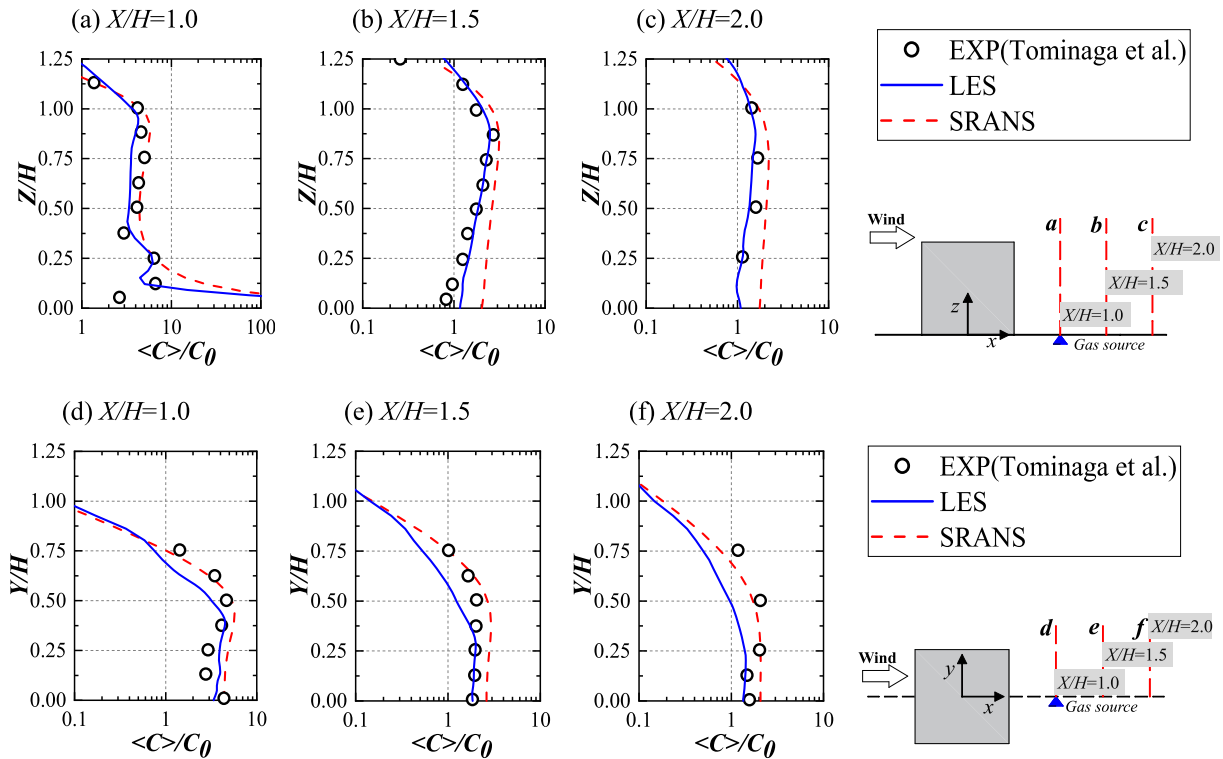


Fig. 10. Profiles of time-averaged dimensionless concentration $\langle C \rangle / C_0$ for light gas behind the building at vertical plane $Y/H = 0$ (a-c) and horizontal plane $Z/H = 0.5$ (d-f).

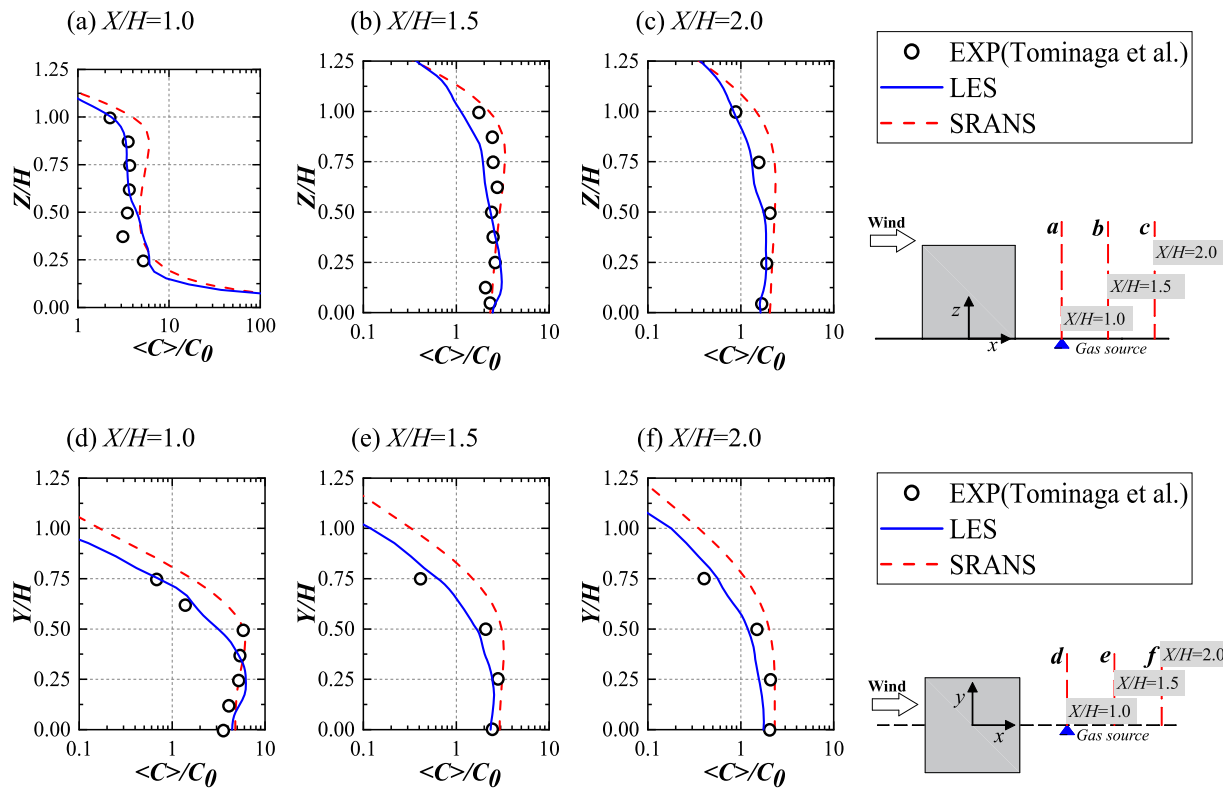


Fig. 11. Profiles of time-averaged dimensionless concentration $\langle C \rangle / C_0$ for neutral gas behind the building at vertical plane $Y/H = 0$ (a–c) and horizontal plane $Z/H = 0.5$ (d–f).

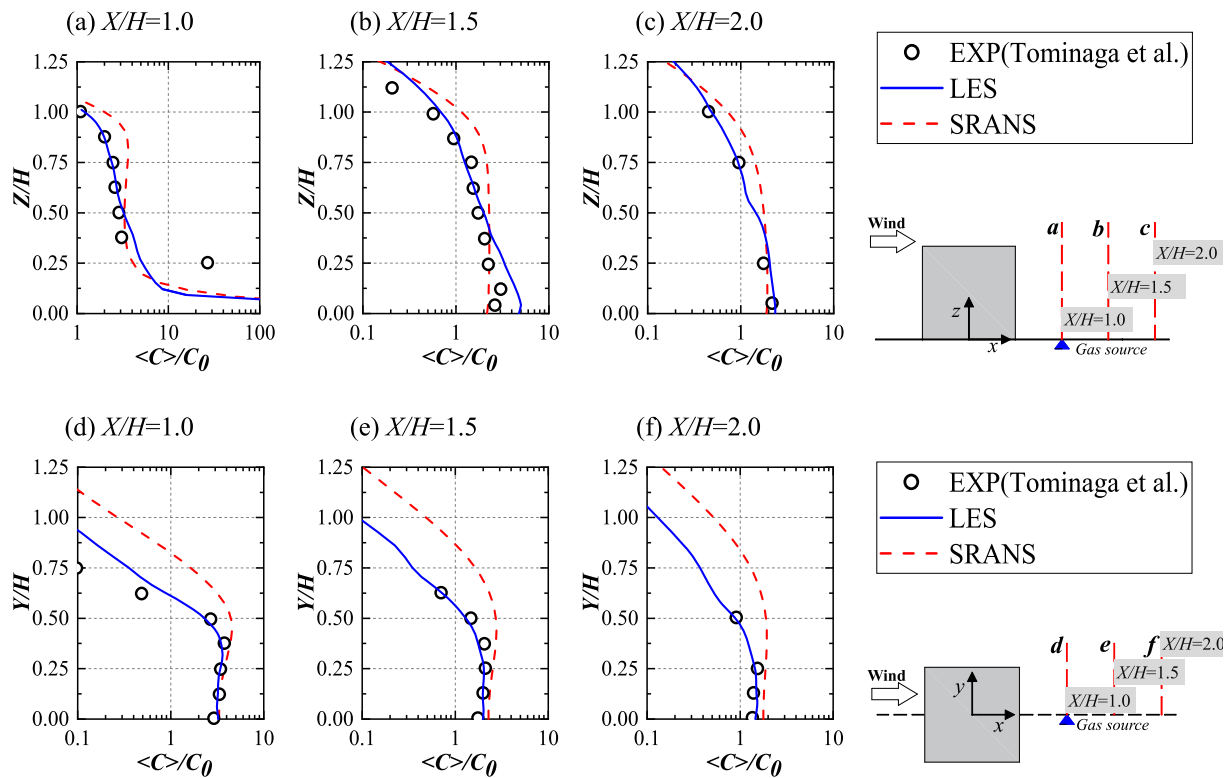


Fig. 12. Profiles of time-averaged dimensionless concentration $\langle C \rangle / C_0$ for heavy gas behind the building at vertical plane $Y/H = 0$ (a–c) and horizontal plane $Z/H = 0.5$ (d–f).

Table 5

Validation metrics for time-averaged dimensionless concentration.

Validation metrics		Vertical plane ($Y/H = 0$)			Horizontal plane ($Z/H = 0.5$)		
		FAC2	FB	NMSE	FAC2	FB	NMSE
Ideal value		1	0	0	1	0	0
Acceptance range		>0.5	$ FB < 0.3$	< 4	>0.5	$ FB < 0.3$	< 4
Light case	LES	0.92	-0.08	0.60	0.79	0.22	0.16
	SRANS	0.83	-0.41	1.30	1.00	-0.19	0.11
Neutral case	LES	1.00	-0.04	0.08	1.00	0.04	0.08
	SRANS	1.00	-0.22	0.10	0.73	-0.16	0.07
Heavy case	LES	0.95	0.27	3.38	0.94	-0.01	0.01
	SRANS	0.86	0.26	3.66	0.76	-0.37	0.23

concentration accumulation level. Therefore, it can be concluded that the positive buoyancy of light gas would benefit the pollutant dilution behind the building, whereas the negative buoyancy of heavy gas has an adverse effect on it, especially in the recirculation region.

5.2. Impact of plume buoyancy on concentration fluxes

The scalar transport of concentration is affected by the combined action of mean flow and turbulent flow, which are usually expressed as convective flux ($\langle u_i \rangle \langle C \rangle$) and turbulent flux ($\langle u_i' C' \rangle$), respectively. Since the distribution of concentration fluxes reflects the essence of contaminant transport [32], it is significant to further illustrate the diffusion mechanism of pollutants with different buoyancies by analyzing the distribution of convective and turbulent fluxes.

Fig. 15 shows the distribution of dimensionless vertical convective and turbulent fluxes at the building height behind the leeward wall. Negative convective flux for the light, neutral, and heavy gas appear here due to the existence of the wake vortex as shown in Fig. 7. However, the role of turbulent flow at the same place is opposite. For the light gas case, the turbulent transport in the recirculation region is much stronger than that of the convective transport in the opposite direction. This indicates that intense turbulent transport helps more light gas break through the limitation of recirculating flow and diffuse upward. In contrast, the difference between the two concentration transport mechanisms for the heavy gas is less significant, which contributes to the accumulation of it at the bottom near the leeward side of the building.

Fig. 16 compares the dimensionless cross-wind convective and turbulent fluxes of three cases behind the building at horizontal plane $Z/H = 0.1$. Similar to the fluxes in vertical direction, due to the existence of the wake vortex behind the building, the effect of horizontal convective flux tends to trap contaminants in the wake region. The turbulent flux, which has an opposite sign to convective flux, tends to transport pollutants across the wind to the outside of the building. A larger turbulent flux is shown in the heavy case, leading to the higher concentration in the cross-wind direction as shown in Fig. 13. The time-averaged

concentration contours at plane $Z/H = 0.1$ are listed in the appendix (Fig. A1), which could better illustrate this phenomenon.

5.3. Impact of plume buoyancy on concentration fluctuations

For hazardous gases that would be detrimental to human health, instantaneous concentration is also the center of interest in addition to time-averaged pollutant distributions. Following the definition of Li and Meroney [60], Fig. 17 illustrates the local concentration fluctuation intensity $I_{C,Loc}$ ($= \sqrt{\langle C'^2 \rangle} / \langle C \rangle$) of different buoyant gases. The local concentration fluctuation intensity $I_{C,Loc}$ is the value of root mean square concentration nondimensionalized by the time-averaged concentration at the same location. Under the effect of different plume buoyancies, a significant difference in terms of the distribution of $I_{C,Loc}$ is observed between the three cases. The $I_{C,Loc}$ appearing above the pollutant source in the light gas case is much larger than that of the neutral gas case, denoting that in this area the concentration fluctuation is more intense than the time-averaged concentration due to the existence of positive buoyancy. The concentration fluctuation of the heavy gas case is weaker than that of the neutral gas case with the high value of $I_{C,Loc}$ distributed downstream behind the pollutant source. It should be noted that the peak value of $I_{C,Loc}$ for the light gas case is larger than 5 while that for the heavy gas case is just around 2. Overall, the positive buoyancy of light gas enhances the concentration fluctuation, whereas the negative buoyancy of heavy gas has an inhibiting effect on it.

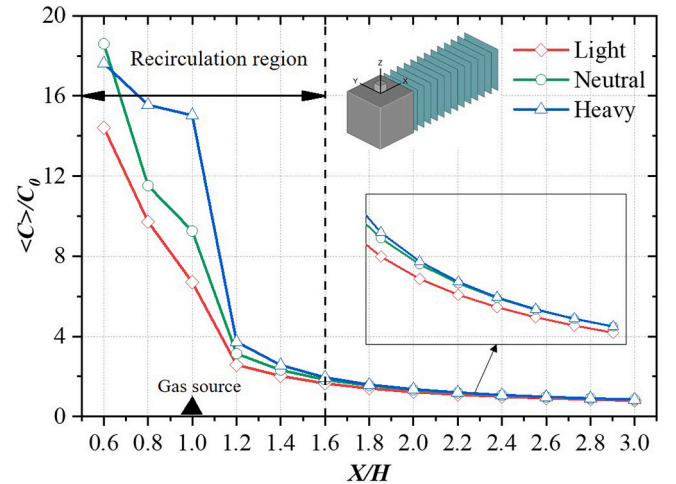


Fig. 14. Plane-averaged dimensionless time-averaged concentration $\langle C \rangle / C_0$ in the wake region.

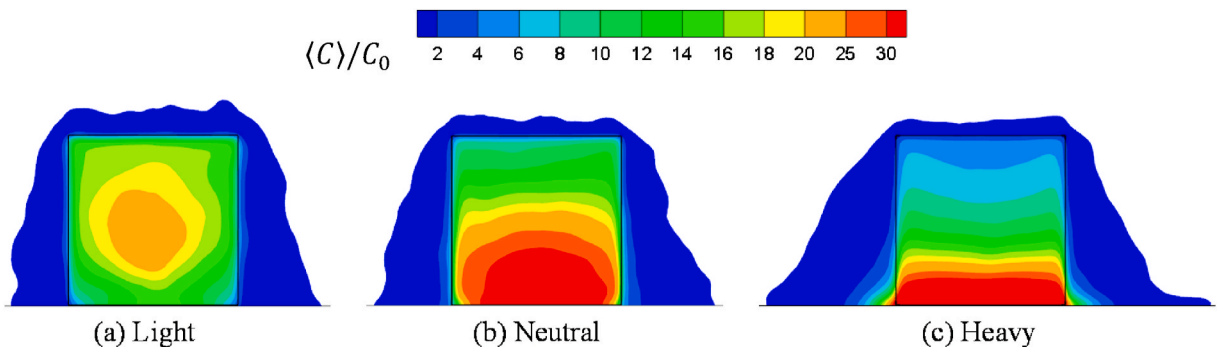


Fig. 13. The distribution of time-averaged dimensionless concentration $\langle C \rangle / C_0$ of pollutants with different densities near the leeward side of the building (plane $X/H = 0.52$).

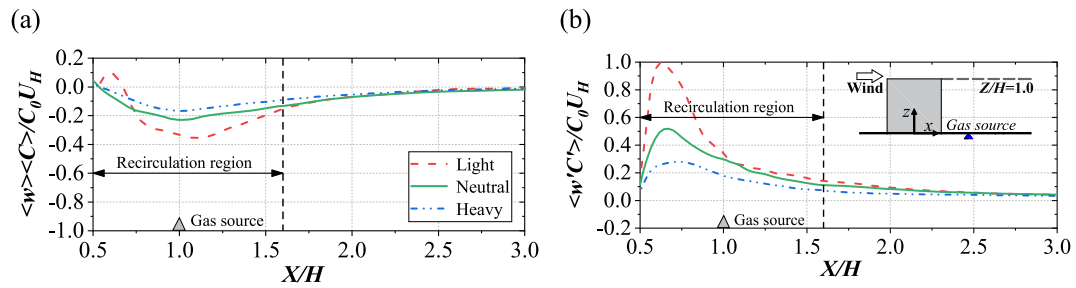


Fig. 15. Comparison of vertical convective flux ($\langle w \rangle \langle C \rangle / C_0 U_H$) (a) and turbulent flux ($\langle w' C' \rangle / C_0 U_H$) (b) for light, neutral and heavy gas at the building height at vertical plane $Y/H = 0$.

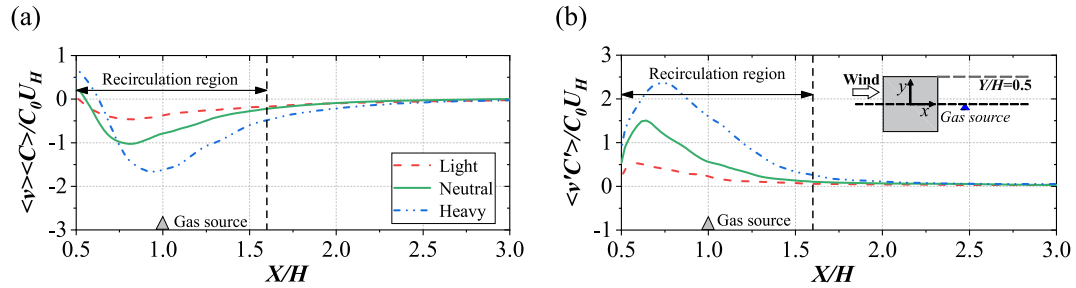


Fig. 16. Comparison of spanwise convective flux ($\langle v \rangle \langle C \rangle / C_0 U_H$) (a) and turbulent flux ($\langle v' C' \rangle / C_0 U_H$) (b) for light, neutral and heavy gas at horizontal plane $Z/H = 0.1$.

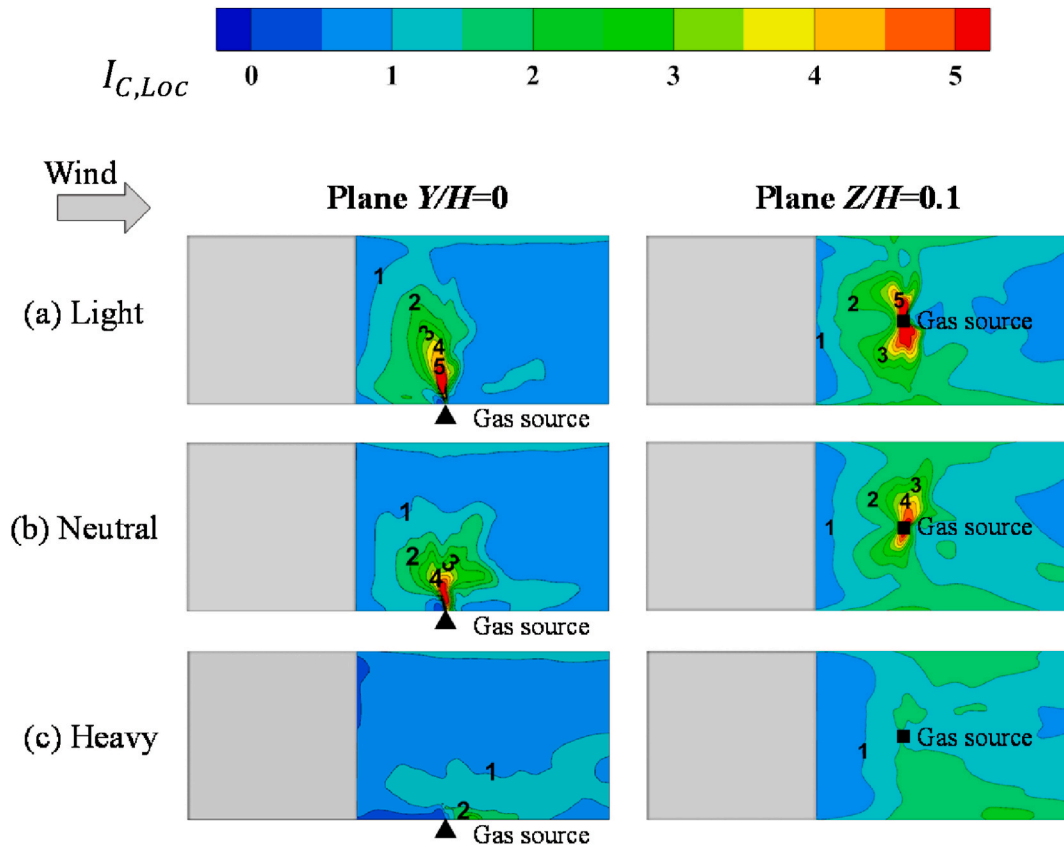


Fig. 17. Contours of local concentration fluctuation intensity ($I_{C,Loc} = \sqrt{\langle C'^2 \rangle} / C$) at vertical plane $Y/H = 0$ and horizontal plane $Z/H = 0.1$.

6. Conclusion

In this paper, numerical simulations were conducted to investigate pollutant dispersion behind the cubic building. Three cases were set to consider the positive and negative buoyancies at the same time, which were produced by the density difference between pollutant gas and ambient air. The predicted results of LES and SRANS, including time-averaged velocity and time-averaged concentration under the effect of different buoyant forces were compared with the experimental data. Meanwhile, the prediction accuracy of the two methods was evaluated. Further researches were subsequently conducted to reveal the influence of plume buoyancies on pollutant dilution behind the building, pollutant transport, and concentration fluctuation.

The main conclusions of the study are summarized as follows:

- (1) Updraft along the leeward side of the building was enhanced by positive buoyancy, while the negative buoyancy had an opposite effect. In addition, the turbulence kinetic energy near the pollution source increased and decreased in light and heavy gas case, respectively.
- (2) Compared with SRANS, LES provided better prediction results for average concentration with different buoyancies, although some underestimations were observed in the cross-wind concentration distribution of the light gas. For gases with different plume buoyancies, LES performed the best for the neutral gas case, with the heavy gas case coming second and the light gas case at last in terms of prediction accuracy.
- (3) After being released from the source, the light gas impinged on the leeward wall of the building directly and raised along it. However, the heavy gas accumulated at the lower part under the effect of negative buoyancy, and diffused largely across the wind. Overall, the positive buoyant force enhanced the dilution of pollutants in the wake region behind the building while the negative buoyancy restrained it.
- (4) The light gas diffused from the top of the wake region mainly under the influence of vertical turbulent flux. However, the upward turbulent transport of heavy gas was inhibited by the

negative buoyancy, which contributed to the accumulation of contaminants in the wake region.

- (5) In cross-wind direction, due to the existence of wake vortex, the convection flux tended to trap pollutant in the wake region, while the turbulent flux tended to transport it outwards. A large cross-wind turbulent flux at the lower part of the building was observed for heavy gas case, leading to its high concentration in the horizontal plane.
- (6) Positive buoyancy of the light gas enhanced the concentration fluctuation between the pollution source and the leeward side of the building, while the negative buoyancy of heavy gas had a certain inhibitory effect on it. For the heavy gas case, high values appear in the lower part behind the source.

In this study, we discussed the characteristics of buoyant gas dispersion with a densimetric Froude number (Fr_d) of $\pm 2.2 \times 10^{-4}$. Only a single value of source volume flux Q_e was considered. In actual situations, Q_e determines the amount of buoyant gases, and therefore determines the buoyant flux. It plays an important role in dispersion characteristics. A small Q_e will result in a near-neutral dispersion, in which the buoyancy effect could be neglected. In further studies, additional efforts, including wind tunnel tests and numerical simulations, are needed to investigate the influence of Q_e .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This project is jointly supported by the National Natural Science Foundation of China (52078380) and the Ministry of Science and Technology of China (SLDRCE19-B-14), which are gratefully acknowledged.

Appendix

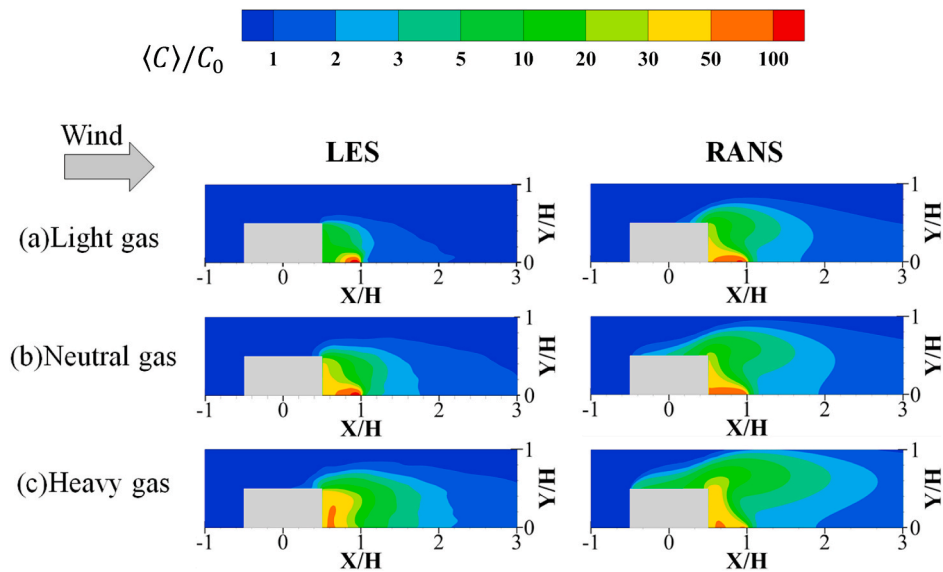


Fig. A.1
Contours of time-averaged dimensionless concentration $\langle C \rangle / C_0$ at horizontal plane $Z/H = 0.1$.

References

- [1] Y. Takano, P. Moonen, On the influence of roof shape on flow and dispersion in an urban street canyon, *J. Wind Eng. Ind. Aerod.* 123 (2013) 107–120.
- [2] H. Kikumoto, R. Ooka, Large-eddy simulation of pollutant dispersion in a cavity at fine grid resolutions, *Build. Environ.* 127 (2018) 127–137.
- [3] G. Jiang, R. Yoshie, Large-eddy simulation of flow and pollutant dispersion in a 3D urban street model located in an unstable boundary layer, *Build. Environ.* 142 (2018) 47–57.
- [4] F. Bazdidi-Tehrani, P. Gholamalipour, M. Kiamansouri, M. Jadidi, Large eddy simulation of thermal stratification effect on convective and turbulent diffusion fluxes concerning gaseous pollutant dispersion around a high-rise model building, *J. Build. Perform. Simulat.* 12 (2018) 97–116.
- [5] Y.D. Huang, M.Z. Li, S.Q. Ren, M.J. Wang, P.Y. Cui, Impacts of Tree-Planting Pattern and Trunk Height on the Airflow and Pollutant Dispersion inside a Street Canyon, vol. 165, *Building and Environment*, 2019.
- [6] D. Cui, X. Li, Y. Du, C.M. Mak, K. Kwok, Effects of Envelope Features on Wind Flow and Pollutant Exposure in Street Canyons, vol. 176, *Building and Environment*, 2020.
- [7] G. Jiang, R. Yoshie, Side Ratio Effects on Flow and Pollutant Dispersion Around an Isolated High-Rise Building in a Turbulent Boundary Layer, vol. 180, *Building and Environment*, 2020.
- [8] X. Zhou, A. Ying, B. Cong, H. Kikumoto, R. Ooka, L. Kang, H. Hu, Large eddy simulation of the effect of unstable thermal stratification on airflow and pollutant dispersion around a rectangular building, *J. Wind Eng. Ind. Aerod.* 211 (2021), <https://doi.org/10.1016/j.jweia.2021.104526>.
- [9] D. Malcangio, M. Mossa, A laboratory investigation into the influence of a rigid vegetation on the evolution of a round turbulent jet discharged within a cross flow, *J. Environ. Manag.* 173 (2016) 105–120.
- [10] R.R. Hwang, T.P. Chiang, W.C. yang, Effect of ambient stratification on buoyant jets in cross-flow, *J. Eng. Mech.* 121 (8) (1995) 865–872.
- [11] C. Cintolesi, A. Petronio, V. Armenio, Turbulent structures of buoyant jet in cross-flow studied through large-eddy simulation, *Environ. Fluid Mech.* 19 (2018) 401–433.
- [12] M. Gao, W. Huai, Y. Xiao, Z. Yang, B. Ji, Large eddy simulation of a vertical buoyant jet in a vegetated channel, *Int. J. Heat Fluid Flow* 70 (2018) 114–124.
- [13] M.F. Reeder, R.E. Huffman, R.D. Branam, K.D. Lebay, S.M. Meents, Near-field development of gas-phase horizontal laminar jets with positive and negative buoyancy measured with filtered Rayleigh scattering, *Exp. Fluids* 50 (2010) 1455–1472.
- [14] R.N. Meroney, Lift off of buoyant gas initially on the ground, *J. Wind Eng. Ind. Aerod.* 5 (1979) 1–11.
- [15] P.T. Roberts, D.J. Hall, Wind tunnel simulation. Boundary layer effects in dense gas dispersion experiments, *J. Loss Prev. Process. Ind.* 7 (1994) 106–117.
- [16] M. Lateb, R.N. Meroney, M. Yataghene, H. Fellouah, F. Saleh, M.C. Boufadel, On the use of numerical modelling for near-field pollutant dispersion in urban environments—A review, *Environ. Pollut.* 208 (2016) 271–283.
- [17] Y. Tominaga, T. Stathopoulos, Ten questions concerning modeling of near-field pollutant dispersion in the built environment, *Build. Environ.* 105 (2016) 390–402.
- [18] A. Robins, Wind tunnel dispersion modelling some recent and not so recent achievements, *J. Wind Eng. Ind. Aerod.* 91 (2003) 1777–1790.
- [19] C. Lin, R. Ooka, H. Kikumoto, T. Sato, M. Arai, Wind tunnel experiment on high-buoyancy gas dispersion around isolated cubic building, *J. Wind Eng. Ind. Aerod.* 202 (2020).
- [20] J. Havens, H. Walker, T.O. Spicer, Wind tunnel study of air entrainment into two-dimensional dense gas plumes at the Chemical Hazards Research Center, *Atmos. Environ.* 35 (2001) 2305–2317.
- [21] A. Robins, I. Castro, P. Hayden, N. Steggel, D. Contini, D. Heist, A wind tunnel study of dense gas dispersion in a neutral boundary layer over a rough surface 35 (2001) 2243–2252.
- [22] A. Robins, I. Castro, P. Hayden, N. Steggel, D. Contini, D. Heist, T.J. Taylor, A wind tunnel study of dense gas dispersion in a stable boundary layer over a rough surface, *Atmos. Environ.* 35 (2001) 2253–2263.
- [23] G.A. Briggs, R.E. Britter, S.R. Hanna, J.A. Havens, A.G. Robins, W.H. Snyder, Dense gas vertical diffusion over rough surfaces results of wind-tunnel studies, *Atmos. Environ.* 35 (2001) 2265–2284.
- [24] B. Blocken, LES over RANS in building simulation for outdoor and indoor applications: a foregone conclusion? *Building Simulation* 11 (2018) 821–870.
- [25] H.A. Olvera, A.R. Choudhuri, W.-W. Li, Effects of plume buoyancy and momentum on the near-wake flow structure and dispersion behind an idealized building, *J. Wind Eng. Ind. Aerod.* 96 (2008) 209–228.
- [26] M.R. Mokhtarzadeh-Dehghan, A. Akcayoglu, A.G. Robins, Numerical study and comparison with experiment of dispersion of a heavier-than-air gas in a simulated neutral atmospheric boundary layer, *J. Wind Eng. Ind. Aerod.* 110 (2012) 10–24.
- [27] O. Han, Y. Zhang, A. Li, J. Li, Y. Li, H. Liu, Experimental and numerical study on heavy gas contaminant dispersion and ventilation design for industrial buildings, *Sustainable Cities and Society* 55 (2020).
- [28] Y. Tominaga, T. Stathopoulos, CFD simulations of near-field pollutant dispersion with different plume buoyancies, *Build. Environ.* 131 (2018) 128–139.
- [29] X. Li, C. Liu, D. Leung, K. Lam, Recent progress in CFD modelling of wind field and pollutant transport in street canyons, *Atmos. Environ.* 40 (2006) 5640–5658.
- [30] T. Tamura, Towards practical use of LES in wind engineering, *J. Wind Eng. Ind. Aerod.* 96 (2008) 1451–1471.
- [31] W. Rodi, Comparison of LES and RANS calculations of the flow around bluff bodies, *J. Wind Eng. Ind. Aerod.* 71 (1997) 55–75.
- [32] Y. Tominaga, T. Stathopoulos, Numerical simulation of dispersion around an isolated cubic building: model evaluation of RANS and LES, *Build. Environ.* 45 (2010) 2231–2239.
- [33] P. Gousseau, B. Blocken, G.J.F. van Heijst, CFD simulation of pollutant dispersion around isolated buildings: on the role of convective and turbulent mass fluxes in the prediction accuracy, *J. Hazard Mater.* 194 (2011) 422–434.
- [34] S.M. Salim, R. Buccolieri, A. Chan, S. Di Sabatino, Numerical simulation of atmospheric pollutant dispersion in an urban street canyon: comparison between RANS and LES, *J. Wind Eng. Ind. Aerod.* 99 (2011) 103–113.
- [35] R. Yoshie, G. Jiang, T. Shirasawa, J. Chung, CFD simulations of gas dispersion around high-rise building in non-isothermal boundary layer, *J. Wind Eng. Ind. Aerod.* 99 (2011) 279–288.
- [36] L.H. Hu, Y. Xu, W. Zhu, L. Wu, F. Tang, K.H. Lu, Large eddy simulation of pollutant gas dispersion with buoyancy ejected from building into an urban street canyon, *J. Hazard Mater.* 192 (2011) 940–948.
- [37] E.M.M. Wingstedt, A.N. Osnes, E. Åkervik, D. Eriksson, B.A.P. Reif, Large-eddy simulation of dense gas dispersion over a simplified urban area, *Atmos. Environ.* 152 (2017) 605–616.
- [38] C. Lin, R. Ooka, H. Kikumoto, T. Sato, M. Arai, CFD Simulations on High-Buoyancy Gas Dispersion in the Wake of an Isolated Cubic Building Using Steady RANS Model and LES, vol. 188, *Building and Environment*, 2021.
- [39] Y. Tominaga, S. Murakami, A. Mochida, CFD prediction of gaseous diffusion around a cubic model using a dynamic mixed SGS model based on composite grid technique, *J. Wind Eng. Ind. Aerod.* 67&68 (1997) 827–841.
- [40] Ansys Inc, Release 18.0, Theory Guide, ANSYS Inc, Canonsburg, PA 15317, USA, 2017.
- [41] J. Smagorinsky, General circulation experiments with the primitive equations: I the basic experiment, *Mon. Weather Rev.* 91 (1963) 99–164.
- [42] Z.T. Ai, C.M. Mak, Large-eddy Simulation of flow and dispersion around an isolated building: analysis of influencing factors, *Comput. Fluid* 118 (2015) 89–100.
- [43] B. Blocken, T. Stathopoulos, J. Carmeliet, CFD simulation of the atmospheric boundary layer: wall function problems, *Atmos. Environ.* 41 (2007) 238–252.
- [44] B.A. Kader, Temperature and concentration profiles in fully turbulent boundary-layers, *Int. J. Heat Mass Tran.* 24 (9) (1981) 1541–1544.
- [45] P.K. Kundu, I.M. Cohen, D.R. Dowling, *Fluid Mechanics*, sixth ed., Academic Press, London, USA, 2016.
- [46] Y. Tominaga, A. Mochidab, R. Yoshiec, H. Kataokad, T. Nozue, M. Yoshikawaf, T. Shirasawac, AIJ guidelines for practical applications of CFD to pedestrian wind environment around buildings, *J. Wind Eng. Ind. Aerod.* 96 (2008) 1749–1761.
- [47] F. Mathey, D. Cokljat, J.P. Bertoglio, E. Sergeant, Assessment of the vortex method for Large Eddy Simulation inlet conditions, *Progress in Computational Fluid Dynamics An International Journal* 6 (2006) 58–67.
- [48] E. Sergeant, Vers une méthodologie de couplage entre la simulation des grandes échelles et les modèles statistiques. PhD thesis, L'Ecole Centrale de Lyon, France, 2002 (in French).
- [49] Z. Liu, Z. Yu, X. Chen, R. Cao, F. Zhu, An Investigation on External Airflow Around Low-Rise Building with Various Roof Types: PIV Measurements and LES Simulations, vol. 169, *Building and Environment*, 2020.
- [50] E. Keshavarzian, R. Jin, K. Dong, K.C.S. Kwok, Effect of Building Cross-Section Shape on Air Pollutant Dispersion Around Buildings, vol. 197, *Building and Environment*, 2021.
- [51] X. Zheng, H. Montazeri, B. Blocken, Large-eddy simulation of pollutant dispersion in generic urban street canyons: guidelines for domain size, *J. Wind Eng. Ind. Aerod.* 211 (2021).
- [52] I.B. Celik, Z.N. Cehreli, I. Yavuz, Index of resolution quality for large eddy simulations, *J. Fluid Eng.* 127 (2005) 949–958.
- [53] S.B. Pope, *Turbulent Flows*, Cambridge University Press, Cambridge, U.K., 2000.
- [54] H. Belkhou, S. Russell, T. Dbouk, M. Mobtil, D. Bougeard, N.-Y. Francois, Large Eddy Simulation of boundary layer transition over an isolated ramp-type micro roughness element, *Int. J. Heat Fluid Flow* 80 (2019).
- [55] C. du Plessis, S.J. Van der Spuy, H.C.R. Reuter, On the effect of the correction of modelled airfoil tonal noise for a spanwise extension, *Aero. Sci. Technol.* 118 (2021).
- [56] Y. Tominaga, A. Mochida, S. Murakami, S. Sawaki, Comparison of various revised k-ε models and LES applied to flow around a high-rise model building with 1:1:2 shape placed within the surface boundary layer, *J. Wind Eng. Ind. Aerod.* 96 (2008) 389–411.
- [57] R.P. Hoxey, P.J. Richards, J.L. Short, A 6 m cube in an atmospheric boundary layer flow Part 1. Full-scale and wind-tunnel results, *Wind Struct.* 5 (2002) 165–176.
- [58] M. Schatzmann, H. Olesen, J. Franke (Eds.), *COST 732 Model Evaluation Case Studies: Approach and Results*, University of Hamburg, Hamburg, Germany, 2010.
- [59] S.R. Hanna, O.R. Hansen, S. Dharmavaram, FLACS CFD air quality model performance evaluation with Kit Fox, MUST, Prairie Grass, and EMU observations, *Atmos. Environ.* 38 (2004) 4675–4687.
- [60] W.W. Li, R.N. Meroney, Gas dispersion near a cubical model building. Part II. Concentration fluctuation measurements, *J. Wind Eng. Ind. Aerod.* 12 (1983) 35–47.